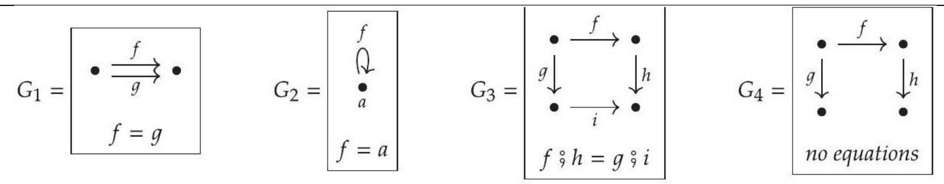
fong-spivak-invitation — formula report

4061 inline formulas (section omitted; 1 did not render), 262 display equations, 68 tables, 505 unrecovered image regions.

Unresolved (2)

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>fong-spivak-invitation_F02349</td><td></td><td>—</td><td><code>\operatorname{cospan} \underline{a} \xrightarrow{ \operatorname{id}_{a}} \underline{a} \stackrel{ \operatorname{id}_{\underline{a}}}{\text { 古 }}</code></td><td>(not rendered)</td><td>—</td></tr>
<tr><td>fong-spivak-invitation_EQ0202</td><td>291</td><td>—</td><td><code>G_{1}=\begin{array}{|c|} \hline \text { 孛 } \\ f=g \end{array} \quad G_{2}=\begin{array}{|c|} \hline f \ \ \stackrel{\mathcal{Q}}{a} \\ f=a \end{array} \quad G_{3}=\begin{array}{|ccc|} \bullet & \xrightarrow{f} & \bullet \\ g \downarrow & & \downarrow \\ \bullet & i \\ f \circ & \bullet \\ f \circ g \circ i \end{array} \quad G_{4}=\begin{array}{|cc|} \bullet & \stackrel{f}{ \longrightarrow} \bullet \\ g \downarrow & \\ \bullet & \bullet \\ \text { no equations } \\ \hline \end{array}</code></td><td>(not rendered)</td><td></td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured</td></tr>
</table>

Flagged, not acted on — 7 of 81 shown

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
fong-spivak-invitation_EQ0134	213	■ 0.235	`\begin{array}{rrrr} \operatorname{in}(\mu) & :=2, & \operatorname{in}(\eta):=0, & \operatorname{in}(\delta):=1, \end{array} \quad \operatorname{in}(\epsilon):=1,,`	$\mathrm{in}(\mu) \quad := 2, \quad \mathrm{in}(\eta) := 0, \quad \mathrm{in}(\delta) := 1, \quad \mathrm{in}(\epsilon) := 1,,$	$\mathit{in}(\mu) := 2, \quad \mathit{in}(\eta) := 0, \quad \mathit{in}(\delta) := 1, \quad \mathit{in}(\epsilon) := 1,$ $\mathit{out}(\mu) := 1, \quad \mathit{out}(\eta) := 1, \quad \mathit{out}(\delta) := 2, \quad \mathit{out}(\epsilon) := 0.$
fong-spivak-invitation_EQ0227	306	■ 0.259	`\begin{aligned} & \bigvee_{a, b, c, d, e \in \mathcal{X}} \alpha^{-1}(x,(a, 1)) \otimes\left(\mathrm{U}_{\mathcal{X}} \times \eta_{\mathcal{X}}\right)((a, 1),(b, c, d)) \\ = & \bigvee_{a, b, c, d, e \in \mathcal{X}} \alpha^{-1}(x,(a, 1)) \otimes \mathrm{U}_{\mathcal{X}}(a, b) \otimes \eta_{\mathcal{X}}(1, c, d)((b, c, d),(1, e)) \otimes \alpha((1, e), y) \\ = & \bigvee_{a, b, c, d, e \in \mathcal{X}} x(x, a) \otimes \epsilon_{\mathcal{X}}(b, c, 1) \otimes \mathrm{U}_{\mathcal{X}}(d, e) \otimes \alpha((1, e), y) \\ = & x(x, y), \end{aligned}`	$\begin{aligned} & \bigvee_{a,b,c,d,e\in\mathcal{X}} \alpha^{-1}(x,(a,1)) \otimes \left(\mathrm{U}_{\mathcal{X}} \times \eta_{\mathcal{X}}\right)((a,1),(b,c,d)) \\ = & \bigvee_{a,b,c,d,e\in\mathcal{X}} \alpha^{-1}(x,(a,1)) \otimes \mathrm{U}_{\mathcal{X}}(a,b) \otimes \eta_{\mathcal{X}}(1,c,d)((b,c,d),(1,e)) \otimes \alpha((1,e),y) \\ = & \bigvee_{a,b,c,d,e\in\mathcal{X}} x(x,a) \otimes \epsilon_{\mathcal{X}}(b,c,1) \otimes \mathrm{U}_{\mathcal{X}}(d,e) \otimes \alpha((1,e),y) \\ = & x(x,y), \end{aligned}$	$\begin{aligned} & \bigvee_{a,b,c,d,e\in\mathcal{X}} \alpha^{-1}(x,(a,1)) \otimes (\mathrm{U}_{\mathcal{X}} \times \eta_{\mathcal{X}})((a,1),(b,c,d)) \\ & \qquad \otimes (\epsilon_{\mathcal{X}} \times \mathrm{U}_{\mathcal{X}})((b,c,d),(1,e)) \otimes \alpha((1,e),y) \\ = & \bigvee_{a,b,c,d,e\in\mathcal{X}} \alpha^{-1}(x,(a,1)) \otimes \mathrm{U}_{\mathcal{X}}(a,b) \otimes \eta_{\mathcal{X}}(1,c,d) \\ & \qquad \otimes \epsilon_{\mathcal{X}}(b,c,1) \otimes \mathrm{U}_{\mathcal{X}}(d,e) \otimes \alpha((1,e),y) \\ = & \bigvee_{a,b,c,d,e\in\mathcal{X}} \mathcal{X}(x,a) \otimes \mathcal{X}(a,b) \otimes \mathcal{X}(c,d) \otimes \mathcal{X}(b,c) \otimes \mathcal{X}(d,e) \otimes \mathcal{X}(e,y) \\ = & \mathcal{X}(x,y), \end{aligned}$
fong-spivak-invitation_EQ0095	146	■ 0.151	`\operatorname{Col}(\Phi)(a, b):= \begin{cases} \mathcal{X}(a, b) & \text { if } a, b \in \mathcal{X} \\ \Phi(a, b) & \text { if } a \in \mathcal{X}, b \in \mathcal{Y} \\ \varnothing & \text { if } a \in \mathcal{Y}, b \in \mathfrak{X} \\ y(a, b) & \text { if } a, b \in \mathcal{Y}\end{cases}`	$\mathrm{Col}(\Phi)(a,b) := \begin{cases} \mathcal{X}(a,b) & \text{if } a,b \in \mathcal{X} \\ \Phi(a,b) & \text{if } a \in \mathcal{X}, b \in \mathcal{Y} \\ \varnothing & \text{if } a \in \mathcal{Y}, b \in \mathfrak{X} \\ y(a,b) & \text{if } a,b \in \mathcal{Y} \end{cases}$	$\mathbf{Col}(\Phi)(a,b) := \begin{cases} \mathcal{X}(a,b) & \text{if } a,b \in \mathcal{X}, \\ \Phi(a,b) & \text{if } a \in \mathcal{X}, b \in \mathcal{Y}, \\ \varnothing & \text{if } a \in \mathcal{Y}, b \in \mathcal{X}, \\ \mathcal{Y}(a,b) & \text{if } a,b \in \mathcal{Y}. \end{cases}$
fong-spivak-invitation_EQ0131	206	■ 0.985	`\operatorname{colim}_{\mathcal{J}} D:=\{(v, d) \mid v \in V \text { and } d \in D(v)\} / \sim`	$\mathrm{colim}_{\mathcal{J}} D := \{(v,d) \mid v \in V \text{ and } d \in D(v)\} / \sim$	$\mathrm{colim}_{\mathcal{J}} D := \big\{(v,d) \mid v \in V \text{ and } d \in D(v)\big\} / \sim$
fong-spivak-invitation_EQ0020	056	■ 0.184	`\begin{array}{rlrl} h_{1}:=\{2 \oslash, 3 \oslash, 10 & , J \bullet, Q \bullet\}, & & i_{1}:=\{4 \bullet, 4 \bullet, 6 \circlearrowright, 6 \diamond, 10 \diamond\}, \\ h_{2}:=\{2 \diamond, 3 \diamond, 4 \diamond, K & , A \end{array},`	$\begin{array}{rlrl} h_1 := \{2\oslash, 3\oslash, 10 & , J\bullet, Q\bullet\}, & & i_1 := \{4\bullet, 4\bullet, 6\circlearrowright, 6\diamond, 10\diamond\}, \\ h_2 := \{2\diamond, 3\diamond, 4\diamond, K & , A \end{array},$	$h_1 := \{2\heartsuit, 3\heartsuit, 10\spadesuit, J\spadesuit, Q\spadesuit\}, \quad i_1 := \{4\clubsuit, 4\spadesuit, 6\heartsuit, 6\diamondsuit, 10\diamondsuit\},$ $h_2 := \{2\diamondsuit, 3\diamondsuit, 4\diamondsuit, K\spadesuit, A\spadesuit\}, \quad i_2 := \{5\spadesuit, 5\heartsuit, 7\heartsuit, J\diamondsuit, Q\diamondsuit\}.$
fong-spivak-invitation_EQ0010	044	■ 0.623	`p \leq g(f(p)) \quad \text { and } \quad f(g(q)) \leq q .`	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$	$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$
fong-spivak-invitation_EQ0143	226	■ 0.754	`\begin{aligned} \circ_{i}: \mathcal{O} & \left(s_{1}, \ldots, s_{m} ; t_{i}\right) \times \mathcal{O}\left(t_{1}, \ldots, t_{n} ; t\right) \\ & \rightarrow \mathcal{O}\left(t_{1}, \ldots, t_{i-1}, s_{1}, \ldots, s_{m}, t_{i+1}, \ldots, t_{n} ; t\right) ; \end{aligned}`	$\begin{aligned} \circ_i : \mathcal{O} & \left(s_1, \ldots, s_m; t_i\right) \times \mathcal{O}\left(t_1, \ldots, t_n; t\right) \\ & \rightarrow \mathcal{O}\left(t_1, \ldots, t_{i-1}, s_1, \ldots, s_m, t_{i+1}, \ldots, t_n; t\right); \end{aligned}$	$\begin{aligned} \circ_i : & \mathcal{O}(s_1, \ldots, s_m; t_i) \times \mathcal{O}(t_1, \ldots, t_n; t) \\ & \rightarrow \mathcal{O}(t_1, \ldots, t_{i-1}, s_1, \ldots, s_m, t_{i+1}, \ldots, t_n; t); \end{aligned}$

Conf. MathPix confidence: ■ $\geq$0.9 ■ 0.5–0.9 ■ $<$0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

The remaining 74 flagged rows are stated as a count.

Each differs from its scan by fewer than 20 components, which is the tail of the distribution rather than its bulk: 42 are component (`C`), 32 weak (`W`). By MathPix confidence: 39 at $\geq$0.9, 13 in 0.6–0.9, 22 below 0.6, 0 with none. Out/465 sampled twenty rows drawn from the $\geq$0.9 band corpus-wide and found eighteen purely typographic differences, one typographic plus a crop overrun, one borderline and *no* unambiguous content errors — so a high-confidence flag here is evidence of a rendering difference, not of a defect. Every row is in `report.ink.json`.